

Introduction to mathematics for economists

Exercise Sheet #4 — Mathcamp 2026, Sciences Po Master's in Economics

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During the tutorials, we will solve the budget-set exercises 1–4, the Cobb–Douglas Kuhn–Tucker question, and the inequality-constrained optimization problem in full. Prioritise these. The unconstrained optimization warm-up is a recommended first step before tackling the Lagrangian and Kuhn–Tucker exercises. A full answer sheet is provided, since we will not have time to solve everything in class.

Theoretical Questions

Consumption Set and Budget Set

Let $X = \mathbb{R}_+^n$ be the **consumption set** of a price-taking economic agent, and let

$$B(p, w) = \{x \in \mathbb{R}_+^n \mid \langle p, x \rangle \leq w\}$$

be his **budget set**, where $p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n$ is a price vector and $w > 0$ is the wealth of the consumer. Recall that $\langle p, x \rangle$ stands for the inner product of p and x in $\mathbb{R}^n$, i.e. $\langle p, x \rangle = \sum_{i=1}^n p_i x_i$.

1. Prove that, for any $\lambda > 0$, we have $B(\lambda p, \lambda w) = B(p, w)$.
2. Prove that $B(p, w)$ is a convex set, i.e. for any $x, y \in B(p, w)$ and for any $\mu \in [0, 1]$, $z = \mu x + (1 - \mu)y \in B(p, w)$.
3. Prove that $B(p, w)$ is a compact set (i.e. bounded and closed). Remember that a subset X of $\mathbb{R}_+^n$ is bounded if there exists $M > 0$ such that for any $x \in X$ we have $\|x\| \leq M$ where $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}_+^n$, i.e. $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$. Moreover, a subset X of $\mathbb{R}_+^n$ is closed if for any converging sequence $(x_m)_{m \in \mathbb{N}}$ of X we have $\lim_{m \rightarrow \infty} x_m = x \in X$.
4. Using questions 2 and 3, explain why the following constrained optimization program always has a solution:

$$\begin{cases} \max_{x \in \mathbb{R}_+^n} u(x) \\ \text{s.t. } \langle p, x \rangle \leq w \end{cases}$$

where $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is a continuous utility function.

5. Suppose there are two goods x_1, x_2 with prices p_1, p_2 and budget w . Assume the economic agent has utility

$$U(x_1, x_2) = \ln(x_1^\alpha x_2^\beta),$$

with $\alpha + \beta < 1$ and $\alpha, \beta > 0$. The agent wants to maximize utility:

$$\max_{x_1 p_1 + x_2 p_2 = w} U(x_1, x_2).$$

- Is this optimization problem parametric or not?
- Express x_2 as a function of x_1 .
- Solve the equivalent unconstrained optimization problem.

Solution

- Let's take an arbitrary $\lambda > 0$. Then:

$$\begin{aligned} B(\lambda p, \lambda w) &= \{x \in \mathbb{R}_+^n \mid \langle \lambda p, x \rangle \leq \lambda w\} = \left\{x \in \mathbb{R}_+^n \mid \sum_{i=1}^n (\lambda p_i) x_i \leq \lambda w\right\} \\ &= \left\{x \in \mathbb{R}_+^n \mid \lambda \left(\sum_{i=1}^n p_i x_i\right) \leq \lambda w\right\} = \left\{x \in \mathbb{R}_+^n \mid \sum_{i=1}^n p_i x_i \leq w\right\} = B(p, w). \end{aligned}$$

Economic reading. This is why we say demand only depends on *relative* prices and *real* wealth: scaling all prices and wealth by the same λ leaves the true opportunity set unchanged.

- Take $x, y \in B(p, w)$ and $\mu \in [0, 1]$ and let $z = \mu x + (1 - \mu)y$. We thus have:

$$\sum_{i=1}^n p_i x_i \leq w \quad \text{and} \quad \sum_{i=1}^n p_i y_i \leq w.$$

Therefore:

$$\mu \left(\sum_{i=1}^n p_i x_i\right) + (1 - \mu) \left(\sum_{i=1}^n p_i y_i\right) \leq \mu w + (1 - \mu)w.$$

Then:

$$\sum_{i=1}^n p_i (\mu x_i + (1 - \mu)y_i) \leq w,$$

so $\sum_{i=1}^n p_i z_i \leq w$. Hence, $z \in B(p, w)$.

- Boundedness:** We know from the definition of $B(p, w)$ that $p_i x_i \leq w$ for every $i \in \{1, \dots, n\}$. Therefore, $x_i < (w/p_i)$ for every i . Thus:

$$\sqrt{\sum_{i=1}^n x_i^2} \leq \sqrt{\sum_{i=1}^n \left(\frac{w}{p_i}\right)^2}.$$

Defining $M = \sqrt{\sum_{i=1}^n \left(\frac{w}{p_i}\right)^2}$ gives $\|x\| \leq M$ for every $x \in B(p, w)$: $B(p, w)$ is bounded.

Closure: Take a sequence $(y_k)_{k \in \mathbb{N}}$ of elements of $B(p, w)$ with $y_k \xrightarrow[k \rightarrow \infty]{} y \in \mathbb{R}_+^n$. We know $\sum_{i=1}^n p_i y_{i,k} \leq w$ for all k . Taking limits,

$$\sum_{i=1}^n p_i \lim_{k \rightarrow \infty} y_{i,k} \leq w \implies \langle p, y \rangle \leq w,$$

so $y \in B(p, w)$: $B(p, w)$ is closed, hence compact.

4. We know from question 3 that $B(p, w)$ is compact for every (p, w) , and u is continuous. Hence, by the Weierstrass theorem, u admits a maximum on $B(p, w)$ for every (p, w) . The solution to this optimization program is denoted $x(p, w)$, the **Marshallian demand** of the consumer. Under general assumptions $x(p, w)$ is a set of bundles, so $x : \mathbb{R}_+^n \times \mathbb{R}_+ \rightrightarrows \mathbb{R}_+^n$ is a *correspondence* rather than a function.^a One needs additional assumptions to guarantee that $x(p, w)$ is a singleton, so that x is a genuine function.
5. (a) This problem is clearly parametric: the optimal quantities x_1 and x_2 depend on prices, wealth, and relative preferences α, β .
 (b) Using the budget constraint $x_1 p_1 + x_2 p_2 = w$:

$$x_2 = \frac{w - x_1 p_1}{p_2}.$$

- (c) We maximize:

$$\begin{aligned} \ln(x_1^\alpha x_2^\beta) &= \ln \left(x_1^\alpha \left(\frac{w - x_1 p_1}{p_2} \right)^\beta \right) \\ &= \alpha \ln(x_1) + \beta \ln(w - x_1 p_1) - \beta \ln(p_2). \end{aligned}$$

Taking the FOC:

$$\begin{aligned} \partial_{x_1} \ln(x_1^\alpha x_2^\beta) &= 0 \\ \Leftrightarrow \frac{\alpha}{x_1} - \beta \frac{1}{w - x_1 p_1} p_1 &= 0 \\ \Leftrightarrow \alpha x_1 p_1 + \beta p_1 x_1 &= \alpha w \\ \Leftrightarrow x_1 p_1 (\alpha + \beta) &= \alpha w \\ \Leftrightarrow x_1 &= \frac{\alpha}{\alpha + \beta} \frac{w}{p_1}. \end{aligned}$$

This gives:

$$x_2 = \frac{\beta}{\alpha + \beta} \frac{w}{p_2}.$$

Concavity ensures this is a maximum. **Why is the function concave?**

^aA correspondence generalizes a function: a function maps an element of a set to a unique element of another set, while a correspondence maps an element to a *set*. See section M.H of the mathematical appendix of Mas-Colell, Whinston and Green.

Optimization

Unconstrained

We consider the function f from $\mathbb{R}^2$ to $\mathbb{R}$, defined by:

$$f(x_1, x_2) = \exp [x_1^2 - x_1 x_2 + x_2^2].$$

1. Compute the gradient of f in $(0, 0)$.
2. Compute the Hessian matrix of f in $(0, 0)$.
3. Show that $(0, 0)$ is a global minimum for f .

Solution

Throughout, write $q(x_1, x_2) = x_1^2 - x_1 x_2 + x_2^2$, so that $f = \exp(q)$, and denote $q_1 = \partial q / \partial x_1 = 2x_1 - x_2$ and $q_2 = \partial q / \partial x_2 = 2x_2 - x_1$.

1. By the chain rule, $\nabla f = \exp(q) \nabla q$, i.e.

$$\nabla f(x_1, x_2) = \exp [x_1^2 - x_1 x_2 + x_2^2] \begin{bmatrix} 2x_1 - x_2 \\ 2x_2 - x_1 \end{bmatrix}.$$

Since $\exp(\cdot) > 0$ always, the gradient vanishes exactly where ∇q does, i.e. where $2x_1 = x_2$ and $2x_2 = x_1$. Hence

$$\nabla f(0, 0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and $(0, 0)$ is the *only* critical point of f .

2. Differentiating once more, and using $q_{11} = q_{22} = 2$, $q_{12} = -1$:

$$H_f = \exp(q) \begin{bmatrix} q_1^2 + 2 & q_1 q_2 - 1 \\ q_1 q_2 - 1 & q_2^2 + 2 \end{bmatrix} = \exp [x_1^2 - x_1 x_2 + x_2^2] \begin{bmatrix} (2x_1 - x_2)^2 + 2 & (2x_1 - x_2)(2x_2 - x_1) - 1 \\ (2x_1 - x_2)(2x_2 - x_1) - 1 & (2x_2 - x_1)^2 + 2 \end{bmatrix}.$$

At the origin $q = 0$ and $q_1 = q_2 = 0$, so

$$H_f(0, 0) = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Its leading principal minors are $D_1 = 2 > 0$ and $D_2 = \det H_f(0, 0) = 2 \cdot 2 - (-1)^2 = 3 > 0$. By Sylvester's criterion $H_f(0, 0)$ is **positive definite**. (Equivalently: its eigenvalues are 1 and 3, both strictly positive.) Combined with $\nabla f(0, 0) = 0$, the second-order *sufficient* condition holds and $(0, 0)$ is a **strict local minimum** of f .

3. Since $f = e^q$, the chain and product rules give, in matrix form,

$$H_f(x) = e^{q(x)} \left(\nabla q(x) \nabla q(x)^\top + H_q \right), \quad H_q = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix},$$

where H_q is *constant* because q is quadratic. Now take any direction $z \neq 0$. Since $z^\top (\nabla q \nabla q^\top) z = (\nabla q \cdot z)^2$ and $z^\top H_q z = 2z_1^2 - 2z_1 z_2 + 2z_2^2 = 2q(z)$,

$$z^\top H_f(x) z = e^{q(x)} \left[(\nabla q(x) \cdot z)^2 + 2q(z) \right] \geq 2e^{q(x)} q(z) > 0$$

by $(\star)$. So H_f is positive definite *at every point* of $\mathbb{R}^2$, i.e. f is **strictly convex**. A strictly convex function has at most one critical point, and any critical point is its unique global minimum — and by question 1 that point is $(0, 0)$.

4. Write $Q(x_1, x_2) = (x_1 - x_2)^2$, so $g = \exp(Q)$, with $Q_1 = 2(x_1 - x_2) = -Q_2$ and $Q_{11} = Q_{22} = 2$, $Q_{12} = -2$. Then

$$\nabla g = e^Q \begin{bmatrix} 2(x_1 - x_2) \\ -2(x_1 - x_2) \end{bmatrix}, \quad \text{so} \quad \nabla g(0, 0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Setting $u = 2(x_1 - x_2)$, the Hessian is

$$H_g = e^Q \begin{bmatrix} u^2 + 2 & -(u^2 + 2) \\ -(u^2 + 2) & u^2 + 2 \end{bmatrix} = e^Q(u^2 + 2) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

whose two columns are proportional: H_g has **rank 1** and $\det H_g = 0$ at every point, in particular $H_g(0,0) = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ with $\det = 0$. This matrix is positive semidefinite but *not* definite, so the second-order test is **inconclusive** and tells us nothing about the nature of $(0,0)$.

We must therefore argue directly, exactly as in the shortcut above: $Q(x) = (x_1 - x_2)^2 \geq 0$, so $g(x) \geq e^0 = 1 = g(0,0)$ and $(0,0)$ is a global minimum. But it is not the only one: $Q(x) = 0$ for every point on the line $x_1 = x_2$, so

$$\arg \min_{x \in \mathbb{R}^2} g = \{(t, t) : t \in \mathbb{R}\},$$

an entire line of global minimisers, all attaining the value 1.

Equality Constraints

Lagrangian with Equality Constraints

To maximize (or minimize) $f(x)$ subject to equality constraints $g_1(x) = 0, \dots, g_m(x) = 0$, we form the **Lagrangian**

$$L(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j g_j(x)$$

and look for critical points $\nabla_x L = 0$, $\nabla_\lambda L = 0$, provided the constraint gradients are linearly independent at the candidate point (*constraint qualification*). The nature of each critical point (max, min, or saddle) is then assessed using the **bordered Hessian**.

We want to find the point of the circle of $\mathbb{R}^3$ defined by

$$x^2 + y^2 + z^2 = 1, \quad x + y + z = 0$$

such that the distance between this point and $(1, 1, 0)$ is maximal.

1. Write the Lagrangian of the problem. (*Hint: you need to define the objective first.*)
2. Compute the critical points, and check that in these points, the equality constraints are qualified (i.e. the gradients of the constraints are non-degenerate/linearly independent).
3. Find the maximum and the minimum.

Solution

1. The Lagrangian is:

$$L(\lambda, \mu, x, y, z) = (x-1)^2 + (y-1)^2 + z^2 + \lambda(1 - x^2 - y^2 - z^2) + \mu(-x - y - z).$$

2. Critical points satisfy:

$$\begin{aligned} \frac{\partial L}{\partial x} &= 2(1-\lambda)x - 2 - \mu = 0, \\ \frac{\partial L}{\partial y} &= 2(1-\lambda)y - 2 - \mu = 0, \\ \frac{\partial L}{\partial z} &= 2(1-\lambda)z - \mu = 0, \\ \frac{\partial L}{\partial \lambda} &= x^2 + y^2 + z^2 = 1, \\ \frac{\partial L}{\partial \mu} &= x + y + z = 0. \end{aligned}$$

Solving this system of nonlinear equations for the 5 unknowns (λ, μ, x, y, z) :

- 1) Summing the first three equations leads to $\mu = -4/3$ (factorizing by $(x+y+z)$ and using $x+y+z=0$).
 - 2) Replacing μ in the first three equations gives $x = y = -z/2$.
 - 3) Replacing x and y by $-z/2$ in $x^2 + y^2 + z^2 = 1$ gives $z = \pm \frac{2}{\sqrt{6}}$, thus $x = y = \pm \frac{1}{\sqrt{6}}$.
 - 4) Replacing x in $2(1-\lambda)x = \frac{2}{3}$ gives $\lambda = 1 \pm \frac{\sqrt{6}}{3}$.
- Solution 1: $(1 + \frac{\sqrt{6}}{3}; -\frac{4}{3}; -\frac{1}{\sqrt{6}}; -\frac{1}{\sqrt{6}}; \frac{2}{\sqrt{6}})$
 - Solution 2: $(1 - \frac{\sqrt{6}}{3}; -\frac{4}{3}; \frac{1}{\sqrt{6}}; \frac{1}{\sqrt{6}}; -\frac{2}{\sqrt{6}})$

The gradients of the constraints at these critical points are $(1, 1, 1)$ and $\pm \frac{1}{6}(1, 1, -2)$, which are linearly independent: both candidates are qualified.

3. We use the bordered Hessian to conclude about the nature of the critical points. The second partial derivatives are

$$\frac{\partial^2 L}{\partial x^2} = \frac{\partial^2 L}{\partial y^2} = \frac{\partial^2 L}{\partial z^2} = 2 - 2\lambda,$$

and all mixed partials are zero. The bordered Hessian is

$$H_b = \begin{bmatrix} 0 & 0 & \frac{\partial g_1}{\partial x} & \frac{\partial g_1}{\partial y} & \frac{\partial g_1}{\partial z} \\ 0 & 0 & \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} & \frac{\partial g_2}{\partial z} \\ \frac{\partial g_1}{\partial x} & \frac{\partial g_2}{\partial x} & \frac{\partial^2 L}{\partial x^2} & 0 & 0 \\ \frac{\partial g_1}{\partial y} & \frac{\partial g_2}{\partial y} & 0 & \frac{\partial^2 L}{\partial y^2} & 0 \\ \frac{\partial g_1}{\partial z} & \frac{\partial g_2}{\partial z} & 0 & 0 & \frac{\partial^2 L}{\partial z^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -2x & -2y & -2z \\ 0 & 0 & -1 & -1 & -1 \\ -2x & -1 & 2-2\lambda & 0 & 0 \\ -2y & -1 & 0 & 2-2\lambda & 0 \\ -2z & -1 & 0 & 0 & 2-2\lambda \end{bmatrix}.$$

With $n = 3$ choice variables and 2 constraints, we only need to check the $3 - 2 = 1$ last principal minor, i.e. the determinant of the full matrix. Evaluating it at each candidate shows that Solution 1 is a **maximum** and Solution 2 is a **minimum**.

Note: here the bordered Hessian is overkill. Why? (Think about the Weierstrass theorem.)

Inequality Constraints

Kuhn–Tucker Conditions

For $\max_x f(x)$ subject to $h_k(x) \leq 0$ for $k = 1, \dots, K$, form

$$L(x, \lambda) = f(x) + \sum_{k=1}^K \lambda_k (-h_k(x)),$$

and the Kuhn–Tucker (KKT) conditions require, at a candidate optimum x^* : stationarity ($\nabla_x L = 0$), primal feasibility ($h_k(x^*) \leq 0$), dual feasibility ($\lambda_k \geq 0$), and **complementary slackness** ($\lambda_k h_k(x^*) = 0$ for each k). When f is concave and the constraint set is convex, these conditions are sufficient, not just necessary, for a global maximum.

We consider the problem

$$\min_{x \geq 0, y \geq 0, x+y \geq 4} (3x^2 + 4xy + 5y^2).$$

1. Show that it admits at least a solution.
2. Consider the problem

$$\max_{x \geq 0, y \geq 0, x+y \geq 4} -(3x^2 + 4xy + 5y^2)$$

and write the Kuhn–Tucker conditions.

3. Show that the constraint $x + y \geq 4$ is binding.
4. Check one qualification constraint.
5. Find the unique solution.

Solution

1. The constraint set of the problem is:

$$h_1(x, y) = -x \leq 0, \quad h_2(x, y) = -y \leq 0, \quad h_3(x, y) = -x - y + 4 \leq 0.$$

We seek the minimum of a strictly convex function (for all $(x, y) \neq (0, 0)$, $(x, y) \begin{pmatrix} 6 & 4 \\ 4 & 10 \end{pmatrix} (x, y)' > 0$) under convex constraints, which yields existence by the course theorem. *Easy check via Sylvester's criterion: the first principal minor is $\det(6) = 6 > 0$, and the second is $\det \begin{pmatrix} 6 & 4 \\ 4 & 10 \end{pmatrix} = 6 \cdot 10 - 4 \cdot 4 = 44 > 0$.*

2. This is only a reformulation of the minimization as a maximization of the opposite of the value function; the constraints are unchanged. The Lagrangian is:

$$L(\lambda_1, \lambda_2, \lambda_3, x, y) = -(3x^2 + 4xy + 5y^2) + \lambda_1 x + \lambda_2 y + \lambda_3 (x + y - 4),$$

with optimality conditions:

$$\text{w.r.t } x : \quad -(6x + 4y) + \lambda_1 + \lambda_3 = 0 \tag{1}$$

$$\text{w.r.t } y : \quad -(4x + 10y) + \lambda_2 + \lambda_3 = 0 \tag{2}$$

$$\lambda_1 \geq 0, \quad x \geq 0, \quad \lambda_1 x = 0 \tag{3}$$

$$\lambda_2 \geq 0, \quad y \geq 0, \quad \lambda_2 y = 0 \tag{4}$$

$$\lambda_3 \geq 0, \quad -x - y + 4 \geq 0, \quad \lambda_3 (-x - y + 4) = 0. \tag{5}$$

3. We identify all possible combinations of binding constraints and rule out infeasible ones: $\emptyset$, $\{h_1\}$, $\{h_2\}$, $\{h_3\}$, $\{h_1, h_2\}$, $\{h_1, h_3\}$, $\{h_2, h_3\}$, $\{h_1, h_2, h_3\}$. The last is ruled out since $h_1 = h_2 = 0$ implies $x = y = 0$, which violates h_3 .

If $\lambda_3 = 0$ (h_3 not binding), stationarity forces neither λ_1 nor λ_2 to be 0, which forces h_1 and h_2 both binding — a contradiction (that case was just ruled out). Alternatively, constructively: suppose for contradiction the optimum (x, y) satisfies $x + y > 4$. Then for small $\varepsilon > 0$ with $\tilde{x} = x - \varepsilon$,

$$f(\tilde{x}, \tilde{y}) - f(x, y) = -[3(x - \varepsilon)^2 + 4(x - \varepsilon)y + 5y^2] + [3x^2 + 4xy + 5y^2] = \varepsilon(6x + 4y - 3\varepsilon).$$

Choosing $\varepsilon = x + y - 4 > 0$ (small enough to keep $\tilde{x} \geq 0$) makes this strictly positive, contradicting optimality of (x, y) . Therefore, at the optimum, $x + y = 4$.

4. When $\{h_1, h_3\}$ is effective, $x = 0$ and $x + y = 4$ so $y = 4$. Then

$$Dh(x, y) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

which has full rank: the qualification constraint is satisfied (and similarly for $\{h_2, h_3\}$). When only h_3 is effective, $Dh(x, y) = \begin{pmatrix} 1 & 1 \end{pmatrix}$ has rank 1, so the qualification constraint holds there too.

5. We examine each set of effective constraints:

- 1) **Case $\{h_3\}$:** $x > 0, y > 0$ imply $\lambda_1 = \lambda_2 = 0$ by complementary slackness. The first two equations give $6x + 4y = \lambda_3$ and $4x + 10y = \lambda_3$. With $x + y = 4$, this yields $x = 3, y = 1, \lambda_3 = 22$, and the value function equals -44 .
- 2) **Case $\{h_2, h_3\}$:** $x > 0, \lambda_1 = 0, y = 0$ gives $x = 4, \lambda_3 = 24, \lambda_2 = -8$ — value -48 , but $\lambda_2 < 0$ violates dual feasibility.
- 3) **Case $\{h_1, h_3\}$:** $y > 0, \lambda_2 = 0, x = 0$ gives $y = 4, \lambda_3 = 40, \lambda_1 = -24$ — value -80 , but $\lambda_1 < 0$ violates dual feasibility.

Only $(x, y) = (3, 1)$ is an admissible KKT point. It is the unique candidate for a global optimum, and by question 1 the global optimum exists — so $(3, 1)$ is the global optimum.

Cobb–Douglas Utility Function

A rational, price-taking consumer has preferences represented by a **Cobb–Douglas utility function**:

$$u(x) = \prod_{i=1}^n x_i^{\alpha_i}, \quad \text{for all } x \in \mathbb{R}_{++}^n, \text{ where } 0 < \alpha_i < 1 \text{ for every } i \text{ and } \sum \alpha_i < 1.$$

Market prices are $p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n$ and the wealth of the consumer is $w > 0$.

1. Justify why the budget constraint must be binding at the optimum.
2. Justify why the optimal solution of the consumer's problem must be unique.
3. Justify why the optimal solution must be interior, i.e. all quantities must be strictly positive at the optimal bundle x^* .
4. Use the Kuhn–Tucker method to solve the consumer's problem and characterize the Marshallian demands $x_i(p, w)$.

Solution

1. For any i , u is strictly increasing in x_i :

$$\forall i, \forall x \in \mathbb{R}_{++}^n, \quad \frac{\partial u}{\partial x_i}(x) = \alpha_i x_i^{\alpha_i-1} \prod_{j \neq i} x_j^{\alpha_j} > 0.$$

Suppose the constraint does not bind at x^* . Then there exists $\varepsilon > 0$ such that $\langle x^* + \varepsilon, p \rangle \leq w$, and since u is strictly increasing the consumer is better off with $x^* + \varepsilon$ — contradicting optimality of x^* .

2. u is strictly concave, continuous, and differentiable on the (convex) budget set, which guarantees a unique maximizer.
3. Whenever $x_i = 0$ for some i , $u(x) = \min_{x \in B(p, w)} u(x) = 0$: a bundle with a zero quantity is never optimal.
4. The Lagrangian is:

$$\mathcal{L}(x, \lambda, \mu) = \prod_{i=1}^n x_i^{\alpha_i} + \lambda(w - \langle p, x \rangle) + \langle \mu, x \rangle.$$

KKT conditions are necessary and sufficient here (concave objective, convex constraints), so the solution (x^*, λ^*, μ^*) is unique and characterized by:

$$\begin{cases} \alpha_i (x_i^*)^{\alpha_i-1} \prod_{j \neq i} (x_j^*)^{\alpha_j} - \lambda^* p_i = 0 & \text{(FOC)} \\ \lambda^*(w - \langle p, x^* \rangle) = 0 & \text{(Compl. slack. BC)} \\ \mu_i^* x_i^* = 0 & \text{(Compl. slack. positivity)} \\ \lambda^* \geq 0, \quad \mu_i^* \geq 0, \quad x_i^* \geq 0. \end{cases}$$

From questions 1 and 3, the budget constraint binds and all $x_i^* > 0$, so by complementary slackness $\lambda^* > 0$ and $\mu_i^* = 0$ for all i . The system reduces to:

$$\begin{cases} \alpha_i (x_i^*)^{\alpha_i-1} \prod_{j \neq i} (x_j^*)^{\alpha_j} = \lambda^* p_i & \text{for all } i \\ \langle p, x^* \rangle = w. \end{cases}$$

For any $i \neq j$, this gives the familiar equality between the MRS and relative prices:

$$\frac{\alpha_i x_j^*}{\alpha_j x_i^*} = \frac{p_i}{p_j} \implies x_j^* = \frac{\alpha_j p_i}{\alpha_i p_j} x_i^*.$$

Substituting this income-expansion path into the budget constraint for each $j \neq i$:

$$p_i x_i^* \left(1 + \sum_{j \neq i} \frac{\alpha_j}{\alpha_i} \right) = w \implies x_i^* = \left(\frac{\alpha_i}{\sum_{j=1}^n \alpha_j} \right) \left(\frac{w}{p_i} \right).$$

So the Marshallian demand for good i is

$$x_i(p, w) = \left(\frac{\alpha_i}{\sum_{j=1}^n \alpha_j} \right) \left(\frac{w}{p_i} \right).$$

Economic reading. Each good's Marshallian demand depends only on its own price and total wealth — not on the other goods' prices — a distinctive (and somewhat unrealistic) feature of Cobb–Douglas preferences that you will revisit when studying cross-price elasticities.

We now assume $X = \mathbb{R}_+^2$, so bundles $x = (x_1, x_2)$ and price vectors $p = (p_1, p_2)$ are two-dimensional.

5. Represent approximately the shape of the budget set $B(p, w)$.

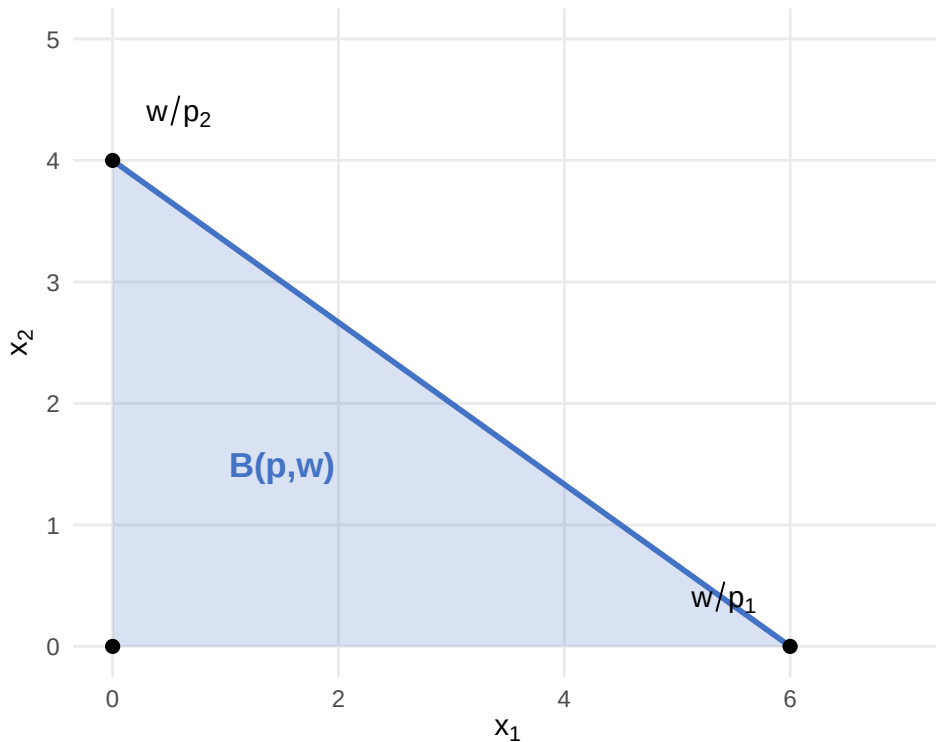


Figure 1

6. Assume now that the State raises a lump-sum tax $T > 0$ on the wealth of the consumer. Represent graphically the effect of this tax on the budget set of the consumer.
7. Suppose instead that the State wants to raise a VAT $t > 0$ on the consumption of good 2. Represent the effect of such a tax on the budget set compared to the initial situation with no tax.
8. How does the tax change the optimal solution?

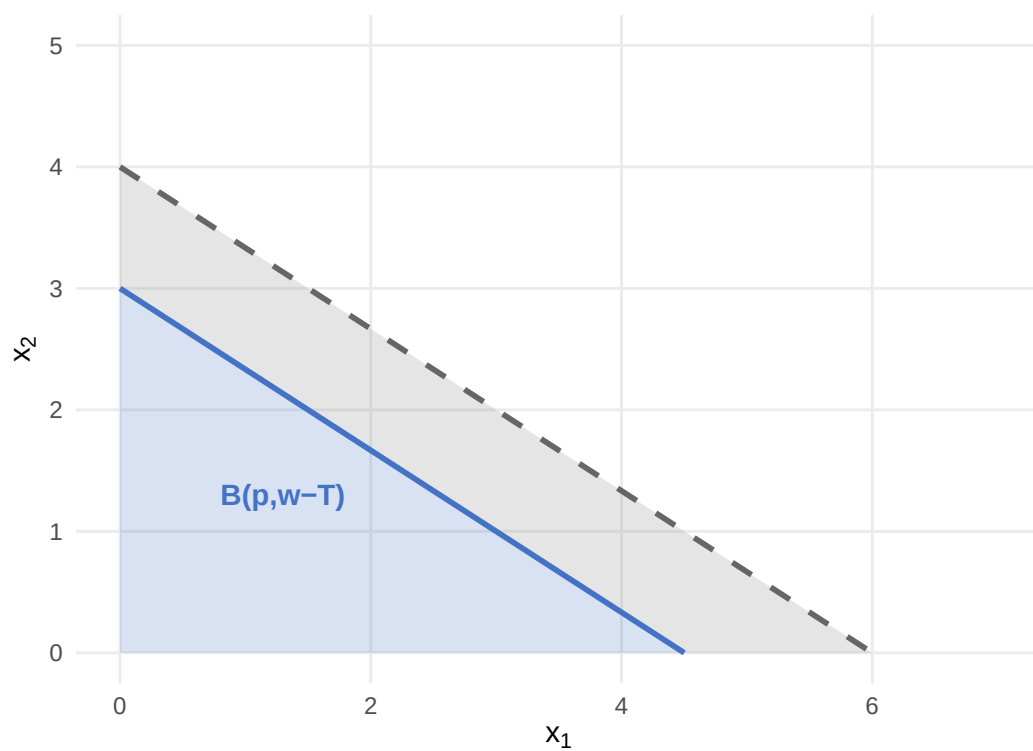


Figure 2

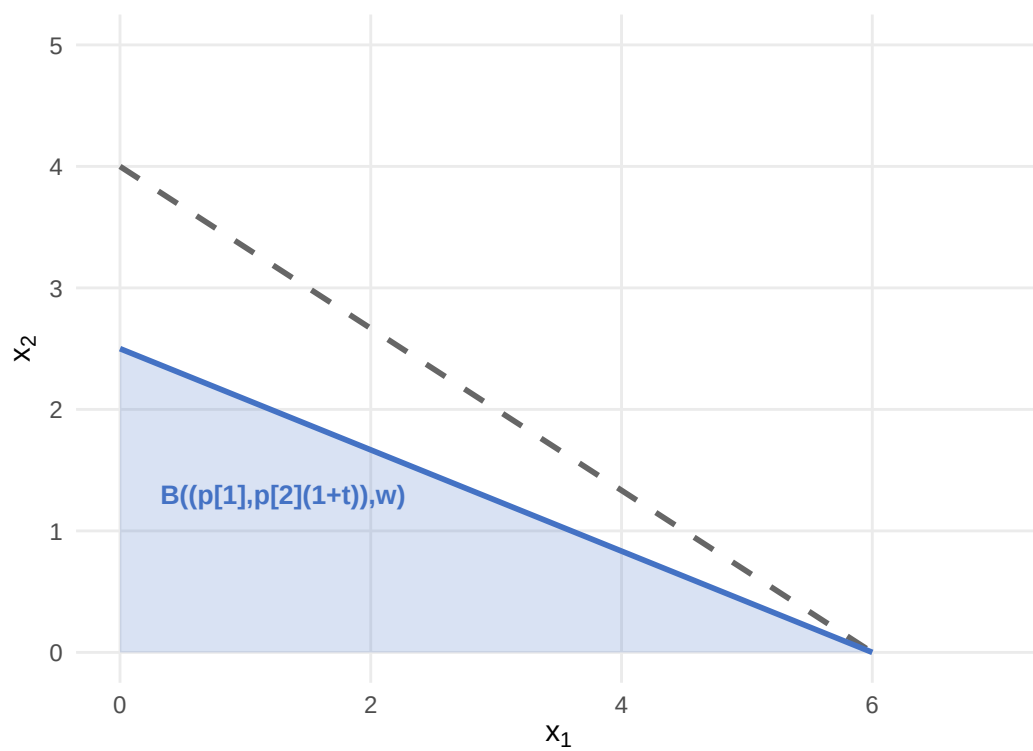


Figure 3

Solution

5. The budget line is $x_2 = -\frac{p_1}{p_2}x_1 + \frac{w}{p_2}$, with intercepts $\left(\frac{w}{p_1}, 0\right)$ and $\left(0, \frac{w}{p_2}\right)$. See the figure above: $B(p, w)$ is the shaded triangle below this line.
6. Wealth becomes $w - T < w$, so the new budget set is $B(p, w - T)$: the budget line shifts *inward, parallel* to the original one — both intercepts shrink by the same proportion $w/(w - T)$.
7. The new budget set is $B((p_1, p_2(1 + t)), w)$: the x_1 -intercept is unchanged (good 1's price is untouched) but the x_2 -intercept shrinks to $\frac{w}{p_2(1 + t)}$, so the budget line pivots — becoming flatter — around the point $(w/p_1, 0)$.
8. Both taxes shrink the feasible set and therefore (weakly) reduce utility at the optimum, but they act differently: the lump-sum tax lowers real income while leaving relative prices — and thus the consumption *ratio* x_1^*/x_2^* — unchanged, whereas the VAT distorts relative prices and induces *substitution* away from good 2 in addition to the income effect. This is the standard argument for why lump-sum taxation is more efficient than distortionary taxation, a theme you will revisit extensively in public economics.

Stone–Geary Utility Function

Stone–Geary Preferences

A rational, price-taking consumer has preferences represented by a **Stone–Geary utility function**:

$$\forall (x_1, x_2) \in \mathbb{R}_+^2, \quad u(x_1, x_2) = \ln(x_1 + \gamma_1) + \ln(x_2 + \gamma_2),$$

where $\gamma_1, \gamma_2 > 0$ are **subsistence parameters**: think of them as minimum quantities the consumer needs before utility becomes meaningfully positive. Market prices are $p = (p_1, p_2) \in \mathbb{R}_{++}^2$ and wealth is $w > 0$.

1. Represent graphically some indifference curves of this utility function. Is it possible that the consumer's problem has corner solutions?

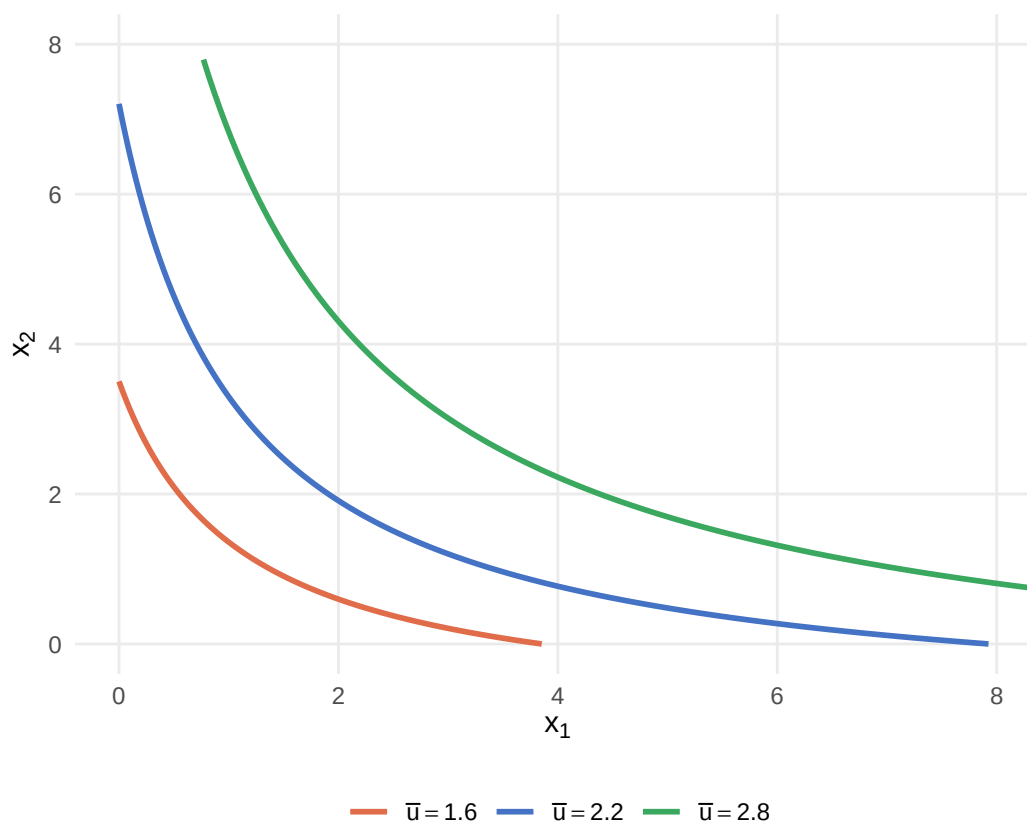


Figure 4

2. Use the Kuhn–Tucker method to solve the consumer's problem and characterize the Marshallian demands.
3. Assume $w + p_2\gamma_2 > p_1\gamma_1$ and $w + p_1\gamma_1 > p_2\gamma_2$. Prove that the Marshallian demand functions are homogeneous of degree zero.

Homogeneous Function

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **homogeneous of degree** $k \in \mathbb{R}$ if, for all $\lambda > 0$ and all $x \in \mathbb{R}^n$,

$$f(\lambda x) = \lambda^k f(x).$$

Solution

1. The indifference curves solve

$$\ln(x_1 + \gamma_1) + \ln(x_2 + \gamma_2) = \bar{u} \Rightarrow x_2 = \exp(\bar{u} - \ln(x_1 + \gamma_1)) - \gamma_2,$$

plotted above for several levels $\bar{u}$. Because each curve can cross an axis at a strictly positive intercept (unlike standard Cobb–Douglas curves, which are asymptotic to the axes), corner solutions *are* possible here — a consumer with low wealth relative to γ_1, γ_2 may optimally consume zero of one good.

2. The Lagrangian is:

$$\forall (x_1, x_2, \lambda, \mu_1, \mu_2) \in \mathbb{R}_+^2 \times \mathbb{R}^3,$$

$$\mathcal{L}(x_1, x_2, \lambda, \mu_1, \mu_2) = \ln(x_1 + \gamma_1) + \ln(x_2 + \gamma_2) + \lambda(w - p_1x_1 - p_2x_2) + \mu_1x_1 + \mu_2x_2.$$

KKT conditions are necessary and sufficient here since u is concave (a sum of concave logs) and the budget constraint is linear, hence convex and continuously differentiable. The optimum is characterized by:

$$\begin{aligned} \frac{1}{x_1^* + \gamma_1} - \lambda^*p_1 + \mu_1^* &= 0 \quad (\text{FOC 1}) \\ \frac{1}{x_2^* + \gamma_2} - \lambda^*p_2 + \mu_2^* &= 0 \quad (\text{FOC 2}) \\ \lambda^*(w - p_1x_1^* - p_2x_2^*) &= 0, \quad \mu_1^*x_1^* = 0, \quad \mu_2^*x_2^* = 0 \\ \lambda^*, \mu_1^*, \mu_2^* &\geq 0, \quad x_1^*, x_2^* \geq 0. \end{aligned}$$

Two simplifications first: (i) since u is strictly increasing, the budget constraint always binds, so we never consider $\lambda^* = 0$; (ii) $\mu_1^* > 0$ and $\mu_2^* > 0$ cannot hold simultaneously, since that would force $x_1^* = x_2^* = 0$, which cannot be optimal given $w, p_1, p_2 > 0$. Three cases remain:

(i) Interior solution ($\lambda^* > 0, \mu_1^* = \mu_2^* = 0$):

$$\begin{cases} \frac{1}{x_1^* + \gamma_1} - \lambda^*p_1 = 0 \\ \frac{1}{x_2^* + \gamma_2} - \lambda^*p_2 = 0 \\ w - p_1x_1^* - p_2x_2^* = 0 \end{cases} \Rightarrow \frac{x_2^* + \gamma_2}{x_1^* + \gamma_1} = \frac{p_1}{p_2}$$

— the familiar MRS-equals-relative-price condition. Combined with the budget constraint, this gives the income-consumption curve $x_2^* = \frac{p_1}{p_2}x_1^* + \frac{p_1\gamma_1 - p_2\gamma_2}{p_2}$, and solving:

$$x_1^* = \frac{w - p_1\gamma_1 + p_2\gamma_2}{2p_1}, \quad x_2^* = \frac{w - p_2\gamma_2 + p_1\gamma_1}{2p_2}, \quad \lambda^* = \frac{2}{w + p_1\gamma_1 + p_2\gamma_2} > 0.$$

This is valid only when $w + p_2\gamma_2 > p_1\gamma_1$ and $w + p_1\gamma_1 > p_2\gamma_2$ (feasibility: $x_1^*, x_2^* > 0$).

(ii) Corner on the x_2 -axis ($x_1^* = 0, \mu_1^* > 0, \mu_2^* = 0$):

$$x_1^* = 0, \quad x_2^* = \frac{w}{p_2} > 0, \quad \lambda^* = \frac{1}{w + p_2\gamma_2} > 0, \quad \mu_1^* = \frac{p_1\gamma_1 - p_2\gamma_2 - w}{\gamma_1(w + p_2\gamma_2)},$$

valid only if $w + p_2\gamma_2 < p_1\gamma_1$ (needed for $\mu_1^* > 0$).

(iii) Corner on the x_1 -axis (symmetric):

$$x_1^* = \frac{w}{p_1}, \quad x_2^* = 0, \quad \lambda^* = \frac{1}{w + p_1\gamma_1}, \quad \mu_2^* = \frac{p_2\gamma_2 - p_1\gamma_1 - w}{\gamma_2(w + p_1\gamma_1)},$$

valid only if $w + p_1\gamma_1 < p_2\gamma_2$.

Putting the three cases together, the Marshallian demands are:

$$x_1(p_1, p_2, w) = \begin{cases} \frac{w - p_1\gamma_1 + p_2\gamma_2}{2p_1} & \text{if } w + p_2\gamma_2 > p_1\gamma_1 \text{ and } w + p_1\gamma_1 > p_2\gamma_2 \\ \frac{w}{p_1} & \text{if } w + p_1\gamma_1 < p_2\gamma_2 \\ 0 & \text{if } w + p_2\gamma_2 < p_1\gamma_1 \end{cases}$$

$$x_2(p_1, p_2, w) = \begin{cases} \frac{w - p_2\gamma_2 + p_1\gamma_1}{2p_2} & \text{if } w + p_2\gamma_2 > p_1\gamma_1 \text{ and } w + p_1\gamma_1 > p_2\gamma_2 \\ 0 & \text{if } w + p_1\gamma_1 < p_2\gamma_2 \\ \frac{w}{p_2} & \text{if } w + p_2\gamma_2 < p_1\gamma_1 \end{cases}$$

Economic reading. Unlike Cobb–Douglas, Stone–Geary demand is *not* simply proportional to w/p_i : the consumer first “sets aside” the subsistence amounts γ_1, γ_2 and only spends the remaining, or *supernumerary*, income according to a Cobb–Douglas-like split. This is the basis of the Linear Expenditure System widely used in applied demand estimation.

3. Let $\lambda > 0$. Then:

$$x_1(\lambda p_1, \lambda p_2, \lambda w) = \frac{\lambda w - \lambda p_1\gamma_1 + \lambda p_2\gamma_2}{2\lambda p_1} = \frac{w - p_1\gamma_1 + p_2\gamma_2}{2p_1} = x_1(p_1, p_2, w),$$

$$x_2(\lambda p_1, \lambda p_2, \lambda w) = \frac{\lambda w - \lambda p_2\gamma_2 + \lambda p_1\gamma_1}{2\lambda p_2} = \frac{w - p_2\gamma_2 + p_1\gamma_1}{2p_2} = x_2(p_1, p_2, w).$$

Marshallian demand functions are homogeneous of degree 0 — exactly the property from Exercise 1.1: only relative prices and real wealth matter, never the units in which money happens to be denominated.

\end{solutionbox}